SagaGen-Academy
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Preprocessing Assignments

You will implement the pipeline using Python and pandas.

1- Read the Dataset

Create the following function

`Read_data_file(file_path)`

The function should read the Titanic csv file and return a panda DataFrame.

Make sure the function can deal with common problems such as:

- The file does not exist.
- The file cannot be read.
- The provided path is invalid.

You do not need to overcomplicate the error handling, but don't assume that the file will always be perfect.

Think about it

What should your function do if the file path is wrong?

Should the program crash with long pandas' error, or should the function provide a more useful message?

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2- Remove Unnecessary Features

Create :

`Drop_unnecessary_features(df, cols_to_drop)`

The function should remove the columns provided through `cols_to_drop`

Important

Don't write something like this inside the function:

`cols_to_drop = ["PassengerId", "Name", "Ticker"]`

the function should not know anything specifically about the Titanic dataset.

Instead, the columns should be come from the configuration.

For example, you may have:

```
Config/
  __init__.py
  Config.py
```

Your configuration can contain the columns that should be removed.

The function should simply receive the list and perform the operation.

Why?

Imagine that tomorrow you use the same function on another dataset.

You should be able to change the configuration without rewriting the function

Inspect the Dataset

Create :

`Check_data_type(df)`

This function should help you understand the structure of the database.

For every column , show at least :

- Column name
- Datatype
- Number of unique values

Return the result as a transpose Data frame that easy to read.

For example, your output should allow you to quickly answer the questions such as :

- What the is Age?
- How many unique values does Embark have?
- Which columns look like categories feature?
- Don't just print `df.dtypes`

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The idea is to create a small data-quality report.

For this structure kindly apply the above two function inside the preprocessing.py file and you can create a main.py file to run the pipeline for read the file and user can input the to remove unnecessary feature saved on config.py also user can check for datatype

```
project/
│
├── config/
│   ├── __init__.py
│   └── config.py
│
├── data/
│   └── raw/
│       └── titanic.csv
│
└── preprocessing.py
 |
 |
 |
 └─ Main.py
```